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Chapter 3

Using All the Information:
Estimating Parameters and Bayes' Theorem

In the last chapter, we described some statistical properties of valued probability distributions, as well as the relationship of samples taken from those distributions to the normal distribution. However, throughout the chapter, we simply assumed in examining the sampling distribution that we knew the parameters of the underlying distribution. But in real life, we often don't know these things. Even for simple things such as coin flips, the "true" distribution of outcomes is that the coin is very slightly biased in one direction or the other. A die has tiny imperfections in its surface or composition that make it more likely to land on one side or another. However, these effects are usually small, and so using the "theoretical" coin which is truly 50-50 is a reasonable approximation for our purposes. Likewise, we generally assume for the purposes of poker analysis that the deck is fair and each card is random until it is observed.

We can mitigate real-world difficulties with distributions that we can reasonably approximate (such as coin flips or die rolls). Other types of distributions, however, pose much more difficult problems. In analyzing poker results, we are often interested in a distribution we discussed in the last chapter - per hand won/loss amounts. When playing in a casino, it would be quite difficult and time-consuming to record the results of every hand - not to mention it might draw unwanted attention. The process is somewhat easier online, as downloadable hand histories and software tools can automate the process. But even if we have all this data, it's just a sample. For poker hands, the probabilities of the underlying distribution won't be reflected in the observed data unless you have a really large amount of data.

We can get around this to some extent by using the subject matter of the last chapter. Suppose that we could get the mean and variance of our hand outcome distribution. Then we could find the sampling distribution and predict the aggregate outcomes from playing different numbers of hands. We can't predict the actual outcome of a particular future sample, but we can predict the distribution of outcomes that will occur.

Now the problem is to try to infer the population mean and variance from the sample mean and variance. We will examine two approaches to this process. The first is the approach of classical statistics, and the second is an approach that utilizes the primary topic of this chapter, Bayes' theorem. The first approach takes as one of its assumptions that we have no information other than the sample about the likelihood of any particular win rate. The second approach postulates a distribution of win rates that exists outside of our particular sample that can be used to refine our estimates of mean and variance for the population distribution.

Estimating Parameters: Classical Statistics

Suppose that we have a player who has played a total of 16,900 hands of limit poker. Normalizing his results to big bets (BB) in order to account for different limits he has played, he has achieved a win rate of $\bar{x}$ = 1.15 BB/100 hands with a standard deviation of s = 2.1 BB/hand. Here instead of μ and σ, which represent population parameters, we use $\bar{x}$ and s, which are sample parameters. Assuming that he plans to continue to play in a similar mix of games with similar lineups of players, what can we say about his "true" win rate μ in the games he has played? We assume in this section that we have no other information about the likelihood of various win rates that might be possible; all win rates from -1 BB/hand to +1BB/hand are deemed to be equally probable.

First of all, it's important to note that we only have a sample to work with. As a result, there will

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be uncertainty surrounding the value of his win rate. However, we know that the sampling distribution of 16,900 hands of limit poker will be approximately normal because of the Central Limit Theorem. The observed standard deviation s = 2.1 BB/h is a reasonable estimate of the standard deviation of the population, particularly because the sample is relatively large. We can use these facts to obtain the ***maximum likelihood estimate*** of the population mean.

Consider all possible win rates. For each of these win rates, there will be a corresponding sampling distribution, a normal curve with a mean of μ and a standard deviation σ_N. The peak of each of these normal curves will be at $x = \mu$, and all the other points will be lower. Now suppose that we assume for a moment that the population mean is in fact some particular μ. The height of the curve at $x = \overline{x}$ will be associated with the probability that the observed rumple mean would have been the result of the sample. We can find this value for all possible values of μ. Since all these normal curves have the same standard deviation σ_N, they will all be identical, but shifted along the X-axis, as in Figure 3.1.

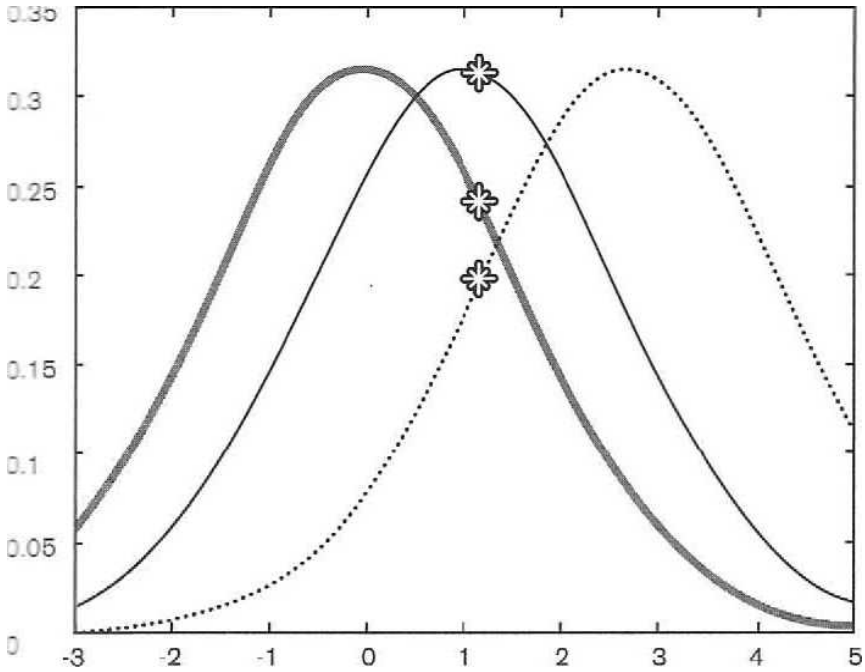

Figure 3.1, Shifted normal distributions (labeled points atx=1.15)

Since the peak of the curve is the highest point, and the observed value $\overline{x}$ is the peak when μ = $\overline{x}$, this means that $\overline{x}$ — 1.15 BB/l00h is the maximum likelihood estimate of the mean of the distribution. This may seem intuitive, but we will see when we consider a different approach to this problem that the maximum likelihood estimate does not always have to equal the sample mean, if we incorporate additional information.

Knowing that the single win rate that is most likely given the sample is the sample mean is a useful piece of data, but it doesn't help much with the uncertainty. After all, our hypothetical player might have gotten lucky or unlucky. We can calculate the standard deviation of a sample of 16,900 hands and we can do some what-if analysis about possible win rates.

Suppose we have a sample N that consists of 16,900 hands taken from an underlying distribution with a mean, or win rate, of 1.15 BB/l00h and a standard deviation of 2.1 BB/h.

Then, using equation 2.4:

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$\sigma_N = \sigma\sqrt{N}$
$\sigma_{16,900} = (2.1\ BB/h)(\sqrt{16{,}900}\ \text{hands})$
$\sigma_{16,900} = 273\ BB$, *so*
$\sigma_N/100h = 273/169 \approx 1.61$

The standard deviation of a sample of this size is greater than the win rate itself. Suppose that we knew that the parameters of the underlying distribution were the same as the observed ones. If we took another sample of 16,900 hands, 32% of the time, the observed outcome of the 16,900 hand sample would be lower than -0.46 BB/100 or higher than 2.76 BB/100.

This is a little troubling. How can we be confident in the idea that the sample represents the true population mean, when even if that were the case, another sample would be outside of even those fairly wide bounds 32% of the time? And what if the true population mean were actually, say, zero? Then 1.15 would fall nicely into the one-sigma Interval. In fact, it seems like we can't tell the difference very clearly based on this sample between a win rate of zero and a win rate of 1.15 BB/100.

What we can do to help to capture this uncertainty is create a ***confidence interval.*** To create a confidence interval, we must first decide on a level of tolerance. Because we're dealing with statistical processes, we can't simply say that the probability that the population mean has some certain value is zero - we might have gotten extremely lucky or extremely unlucky. However, we can choose what is called a ***significance level.*** This is a probability value that represents our tolerance for error, Then the confidence interval is the answer to the question, "What are all the population mean values such that the probability of the observed outcome occurring is less than the chosen significance level?"

Suppose that for our observed player, we choose a significance level of 95%. Then we can find a confidence level for our player. If our population mean is μ, then a sample of this size taken from this population will be between $(\mu - 2\sigma)$ and $(\mu + 2\sigma)$ 95% of the time. So we can find all the values of μ such that the observed value $\bar{x} = 1.15$ is between these two boundaries.

As we calculated above, the standard deviation of a sample of 16,900 hands is 1.61 units/100 hands:

$\sigma_N = \sigma\sqrt{N}$
$\sigma = (2.1\ \text{BB/h})(\sqrt{16{,}900})$
$\sigma = 273$ BB per 16,900 hands
$\sigma/100h = 273$ BB/169 = 1.61

So as long as the population mean satisfies the following two equations, it will be within the confidence interval:

$(\mu - 2\sigma) < 1.15$
$(\mu + 2\sigma) > 1.15$
$(\mu - 2\sigma) < 1.15$
$\mu - (2)(1.61) < 1.15$
$\mu < 4.37$
$(\mu + 2\sigma) > 1.15$
$\mu + (2)(1.61) > 1.15$

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$\mu > -2.07$
So a 95% confidence interval for this player's win rate (based on the 16,900 hand sample he has collected) is [-2.07 BB/100, 4.37 BB/100].

This does not mean that his true rate is 95% likely to lie on this interval. This is a common misunderstanding of the definition of confidence intervals. The confidence interval is all values that, if they were the true rate, then the observed rate would be inside the range of values that would occur 95% of the time. Classical statistics doesn't make probability estimates of parameter values - in fact, the classical view is that the true win rate is either in the interval or it isn't, because it is not the result of a random event. No amount of sampling can make us sure or unsure as to what the parameter value is. Instead, we can only make claims about the likelihood or unlikelihood that we would have observed particular outcomes if a parameter had a particular value.

The maximum likelihood estimate and a confidence interval are useful tools for evaluating what information we can gain from a sample. In this case, even though a rate of 1.15 BB/100 might look reasonable, concluding that this rate is close to the true one is a bit premature. The confidence interval can give us an idea of how wide the range of possible population rates might be. However, if pressed, the best single estimate of the overall win rate is the maximum likelihood estimate, which is the sample mean of 1.15 BB/100 in this case.

To this point, we have assumed that we had no information about the likelihood of different win rates - that is, that our experimental evidence was the only source of data about what win rate a player might have. But in truth, some win rates are likelier than others, even before we take a measurement. Suppose that you played, say, 5,000 hands of casino poker and in those hands you won 500 big bets, a rate of 10 big bets per 100 hands. In this case, the maximum likelihood estimate from the last section would be that your win rate was exactly that -10 big bets per 100 hands,

But we do have other information. We have a fairly large amount of evidence, both anecdotal and culled from hand history databases and the like that indicates that among players who play a statistically significant number of hands, the highest win rates are near 3-4 BB/100. Even the few outliers who have win rates higher than this do not approach a rate of 10 BB/100. Since this is information that we have before we even start measuring anything, we call it a priori information.

In fact, if we just consider any particular player we measure to be randomly selected from the universe of all poker players, there is a probability distribution associated with his win rate. We don't know the precise shape of this distribution, of course, because we lack observed evidence of the entire universe of poker players. However, if we can make correct assumptions about the underlying *a priori* distribution of win rates, we can make better estimates of the parameters based on the observed evidence by combining the two sets of evidence.

Bayes' theorem

In Chapter 2, we stated the basic principle of probability (equation 1.5).

$$p(A \cap B) = p(A)p(B|A)$$

In this form, this equation allows us to calculate the joint probability of A and B from the probability of A and the conditional probability of B given A. However, in poker, we are often most concerned with calculating the conditional probability of B given that A has already-occurred - for example, we know the cards in our own hand (A), and now we want to know how

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this information affects the cards in our opponents hand *(B)*. What we are looking for is the conditional probability of *B* given *A*.

So we can reorganize equation 1.5 to create a formula for conditional probability.

This is the equation we will refer to as ***Bayes' theorem:***

$$p(B|A) = \frac{p(A \cap B)}{p(A)} \qquad \textbf{(3.1)}$$

Recall that we defined $\overline{B}$ as the complement of B in Chapter 1; that is:

$p(\overline{B}) = 1 - p(B)$
$p(B) + p(\overline{B}) = 1$

We already have the definitions:

$$p(A \cap B) = p(A)p(B|A)$$

Since we know that *B* and $\overline{B}$ sum to 1, *p(A)* can be expressed as the probability of *A* given *B* when *B* occurs, plus the probability of *A* given *B* when *B* occurs.

So we can restate equation 3.1 as:

$$p(B|A) = \frac{p(A|B)p(B)}{p(A|B)p(B) + p(A|\overline{B})p(\overline{B})} \qquad \textbf{(3.2)}$$

In poker, Bayes' theorem allows us to refine our judgments about probabilities based on new information that we observe. In fact, strong players use Bayes' theorem constantly as new information appears to continually refine their probability assessments; the process of Bayesian inference is at the heart of reading hands and exploitive play, as we shall see in Part II.

A classic example of Bayes' theorem comes from the medical profession. Suppose that we have a screening process for a particular condition. If an individual with the condition *is* screened, the screening process properly identifies the condition 80% of the time. If an individual without the condition is screened, the screening process improperly identifies him as having the condition 10% of the time. 5% of the population (on average) has the condition.

Suppose, then, that a person selected randomly from the population is screened, and the screening returns a positive result. What is the probability that this person has the condition (absent further screening or diagnosis)?

If you answered "about or a little less than 80%," you were wrong, but not alone. Studies of doctors' responses to questions such as these have shown a perhaps frightening lack of understanding of Bayesian concepts.

We can use Bayes' theorem to find the answer to this problem as follows:
A = the screening process returns a positive result.
B = the patient has the condition.

Then we are looking for the probability $p(B|A)$ and the following are true:
$p(B|A)) = 0.8$

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(if the patient has the condition, the screening will be positive 80% of the time)
$p(A|\overline{B})$ = 0.1

(if the patient doesn't have the condition, the result will be positive 10% of the time)
p(B) = 0.05 (5% of all people have the condition)
$p[\overline{B})$ = 0.95 (95% of all people don't have the condition)

And using Equation 3.2, we have:

$$p(B|A) = \frac{p(A|B)p(B)}{p(A|B)p(B) + p(A|\overline{B})p(\overline{B})}$$

$$p(B|A) = \frac{(0.8)(0.05)}{(0.8)(0.05) + (0.1)(0.95)}$$

$p(B|A) \approx$ 29.63%

As you can see, the probability that a patient with a positive screening result actually has the condition is much lower than the 80% "accuracy" the test has in identifying the condition in someone who has it. Testing with this screening process would need to be followed up by additional testing to confirm the presence of the condition.

But rather than just plugging into an equation, let's discuss what's happening here by considering a population of *100,000* people, all of whom are tested using this screening process.

Of these 100,000 people:
5,000 actually have the condition. (5% of the population)
95,000 actually do not have the condition. (95% of the population)

Of the *5,000* who have the condition:
4,000 will test positive. (80% of those with the condition)
1,000 will test negative. (20% false negatives)

Of the 95,000 who have the condition:
9,500 will test positive. (10% false positives)
85,500 will test negative. (90% of those without the condition)

Now our question was: given that a person has tested positive, how likely are they to have the condition? Out of our 100,000 tested individuals, a total of 13,500 tested positive. Of those, only 4,000 actually have the condition.

$p(B|A)$= 4,000/13,500
$p(B|A) \approx$ 29.6%

Additionally, we can see that by increasing the accuracy of the test, either by making it more accurate in identifying the condition in those who have it, or in producing fewer false positive identifications of the condition in those who do not, we can increase the conditional probability that someone who tests positive has the condition. Note that this does not increase the chance that somebody actually has the condition - we would not want to increase that! - but rather decreases the number of people who incorrectly test positive, and potentially have to incur the stress of incorrectly believing they have a condition.

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Suppose that we increase the effectiveness of the test to always identify those who have the condition, while the false positive rate remains the same. Then, using Bayes' theorem:

$$p(B|A) = \frac{p(A|B)p(B)}{p(A|B)p(B) + p\left(A\middle|\overline{B}\right)p\left(\overline{B}\right)}$$

$$p(B|A) = \frac{(1)(0.05)}{(1)(0.05) + (0.1)(0.95)}$$

$$p(B|A) \approx 34.5\%$$

Likewise, if we hold the 80% identification rate constant and drop the rate of false positives from 10% to 6%, we obtain:

$$p(B|A) = \frac{(0.8)(0.05)}{(0.8)(0.05) + (0.06)(0.95)}$$

$$p(B|A) \approx 41.2\%$$

The key to the application of Bayes' theorem is the existence of a prior probability and obtaining new information. In the above example, we began with a prior probability that our patient had the condition of 5%. After the screening, the positive result was new information that allowed us to revise the probability for the patient - in this case upward. This process is called ***Bayesian inference*** and is critical to successful poker play.

There are countless ways in which Bayesian inference can be used; in fact, many players employ this process unconsciously all the time. Consider a situation in a tournament when a player is at a table that breaks very early on. He is moved to a new table where the player on his right has already amassed a stack seven times as large as the starting stacks, while most of *the* other players in the tournament are still near their starting stacks. Most players would conclude that two scenarios are likely: the player with the huge stack is either an extremely loose and aggressive player or he happens to simply be an ordinary player who got extremely lucky. The natural inclination to believe that he is more likely to be loose and aggressive than very lucky is a direct consequence of Bayes' theorem, whether the player from the broken table is aware of this concept or not.

We can quantify this effect using the formal version of Bayes' theorem, and this can lead us to making stronger plays even without solid information. Consider the following common situation:

A new player sits down in the game. Using all the observational information available to us, which might include stereotyping the player's ethnicity, gender, manner, wardrobe, personal appearance, and so on, we conclude that he is 10% likely to be a "maniac" who will raise 80% of his hands from the cutoff and 90% likely to be a tight player who will raise 10% of his hands from that seat. On the first hand he plays, he raises from the cutoff. (Assume there is no posting.) Now what is the probability- that he is a maniac?

We can use Bayes' theorem to solve this problem, but we invite you to estimate this probability before continuing. We believe that testing intuition is one of the best ways to refine our ability to estimate accurately, a skill chat is invaluable not only in poker but in life.

A = The opponent will raise the first hand he plays from the cutoff.

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B= The opponent is a maniac.

$p(A\|B)$ = 0.8	(if the player is a maniac, he will raise 80% of the time)
$p(A\|\overline{B})$ = 0.1	(if the player is not a maniac, he will raise 10% of the time)
$p(B)$ = 0.1	(10% of the time, he is a maniac *a priori)*
$p(\overline{B})$= 0.9	(90% of the time, he is not a maniac *a priori)*

Applying Bayer's theorem again:

$$p(B|A) = \frac{p(A|B)p(B)}{p(A|B)p(B) + p(A|\overline{B})p(\overline{B})}$$

$$p(B|A) = \frac{(0.8)(0.1)}{(0.8)(0.1) + (0.1)(0.9)}$$

$$p(B|A) \approx 47.1\%$$

So simply by observing this player raising the first hand, we can adjust the probability of this slayer being a maniac from just 10% to 47% immediately. If the player raises the first two hands (assuming the same inference for the seat next to the cutoff], this probability moves to nearly 87%! Of course, these probabilities are subject to the accuracy of our original assumptions - in reality, there are not just two types of players, and our probability estimates are probably not so crisp about what type of player he is.

One tendency among players is to delay characterizing and adjusting to a player's play until gaining a little more information, by observing some hands or the like. But this view is overly passive in our view; maximizing EV means taking advantage of all the information we have at our disposal and not necessarily waiting for confirmation that the information is reliable before trying to take advantage of it. The error that these players are making is that they- do not realize the power of the information they have gained. It is worth noting that many players, even players who do not play well often make this adjustment, or a similar one, intuitively. But beware! This adjustment is open to exploitation by players who will sit down in a game and play very differently from their usual style in an attempt to induce significant adjustments by players in the game.

Strong players use Bayes' theorem constantly as new information appears to continually refine their probability assessments; the process of Bayesian inference is at the heart of reading hands and exploitive play, as we shall see in Part II. But even away from the table, Bayes' theorem can allow us to make more informed conclusions about data. To see an example of this, we return to the topic of win rates.

Estimating Parameters: Bayesian Statistics

Recall earlier in this chapter we discussed a player who had played a total of 16,900 hands of limit poker with the following observed statistics:
Win rate of $\bar{x}$ = 1.15 BB/100 hands
Standard deviation of s = 2.1 BB/hand.

We were concerned with some of the statements that we could make about his "true" win rate based on these observations. Using methods from classical statistics, we found that his maximum likelihood estimator for win rate was 1.15 BB/100 hands and his 95% confidence interval was [-2.07 BB/100, 4.37 BB/100]. These statements relied on an assumption that we had no additional

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information about the distribution of this player's win rates.

However, suppose that we guess at a distribution of win rates and try to apply Bayes' theorem to the distribution of win rates and this player's results in order to produce a more accurate estimate. To do this, we must first hypothesize an overall distribution of win rates for this player. Let's assume that he is pulled randomly from the population of regular poker players. What is the shape of this distribution, which is called a ***prior probability distribution?***

It's pretty unclear at a glance what this distribution looks like - after all, we do not have access to the results and records of the broad population. But we can simplify and estimate, hoping that our distribution of win rates will be close enough to the truth that we can gain from incorporating it into our estimates. Assuming that our player plays some mixture of lower mid-limit games such as $10-$20 to $30- $60, we can estimate the total rake paid by the game as about $3-$4 per hand, or perhaps 0.1 BB/h. Dividing this amongst all the players approximately equally, we obtain a net rake effect on everyone's win rate of about 0.01 BB/h, or 1 BB/100.

The mean of the distribution of all players' win rates, then, is equal to this value, as this is the net flow of money out of the game. Suppose that we have a roughly normal distribution of win rates, let's just estimate a standard deviation (of win rates) of about 0.015 BB/h. This would lead to a population where 68% of players have a rate between -2.5 BB/100 and +0.5 BB/100 and where 95% of players would have a rate between -4 bb/100 and +2 BB/100. This might square with your intuition - if not, these numbers can be tweaked to reflect different assumptions without changing the underlying process.

To simplify the computation, instead of using the continuous normal distribution, we will create a discrete distribution that roughly mirrors our assumptions. We assume that the underlying distribution of all poker players is as follows:

Win Rate	**% of players with this win rate**
-5 BB/100	**0.25%**
-4 BB/100	**2%**
-3 BB/100	**8%**
-2 BB/100	**20%**
-1 BB/100	**39.5%**
0 BB/100	**20%**
+1 BB/100	**8%**
+2 BB/100	**2%**
+3 BB/100	**0.25%**

Now that we have an *a priori* distribution of win rates, we can apply Bayes' theorem to this problem. For each win rate, we calculate:
A = the chance of a win rate of 1.15 being observed.
B = the chance that this particular win rate is the true one *(a priori).*

We cannot directly find the probability of a particular win rate being observed (because the normal is a continuous distribution). We will instead substitute the probability of a win rate between 1.14 and 1.16 being observed as a proxy for this value. Recall that the standard

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deviation of a sample of this size was 1.61 bets. $p(A|\overline{B})$ is calculated by simply calculating the weighted mean of $p(A|B)$ excluding the current row. From a probability chart:

Win Rate	$p(\overline{B})$	$p(A\|B)$	$p(A\|\overline{B})$	$p(B)$
- 5 BB/100	0.25%	0.000004	0.00222555	99.75%
- 4 BB/100	2%	0.000031	0.00226468	98%
- 3 BB/100	8%	0.000182	0.00239720	92%
- 2 BB/100	20%	0.000738	0.00259053	80%
- 1 BB/100	39.5%	0.002037	0.00233943	60.5%
0 BB/100	20%	0.003834	0.00181659	80%
- 1 BB/100	8%	0.004918	0.00198539	92%
- 2 BB/100	2%	0.004301	0.00217753	98%
- 3 BB/100	0.25%	0.002564	0.00221914	99.75%

Applying Bayes' theorem to each of these rows:

$$p(B|A) = \frac{p(A|B)p(B)}{p(A|B)p(B) + p(A|\overline{B})p(\overline{B})}$$

Win Rate	$p(A\|B)$
- 5 BB/100	0.00%
- 4 BB/100	0.03%
- 3 BB/100	0.66%
- 2 BB/100	6.65%
- 1 BB/100	36.25%
0 BB/100	34.54%
- 1 BB/100	17.72%
- 2 BB/100	3.87%
- 3 BB/100	0.29%
Total	100%

When we looked at the classical approach, we were able to generate a maximum likelihood estimator. In the same way, we can identify -1 BB/100 as the maximum likelihood estimate given these assumptions. Of course, these assumptions aren't really the truth - possible win rates for players are close to continuous. Nevertheless, if we used a continuous distribution and did the more complex math that arises, we would find a distribution similar to this one. And the key implication of this approach is that because of the relative scarcity of winning players, our hypothetical hero is nearly as likely to be a losing player who has gotten lucky as he is to have a positive win rate.

We could see this even more starkly if we considered a player with a much higher observed win rate, perhaps 5 BB/l00h. The classical approach would still assign a maximum likelihood estimate of 5 BB/l00h, because it considers all win rates to be equally likely (because in the classical method we have no additional information about the distribution of win rates). However, recalculating the above analysis with an observed win rate of 5 BB. we find:

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Win Rate	$p(A\|B)$
- 5 BB/100	0.00%
- 4 BB/100	0.00%
- 3 BB/100	0.00%
- 2 BB/100	0.16%
- 1 BB/100	3.79%
0 BB/100	15.78%
- 1 BB/100	35.40%
- 2 BB/100	33.84%
- 3 BB/100	11.03%
Total	100%

We can see that here, our player is a heavy favorite to be a winning player, and a substantial one at that. However, as win rates of 5 BB/100 hands are absent from the population, Bayes' theorem properly adjusts his win rate to the more likely levels of the top 10% of all players.

Gathering more data should naturally cause these Bayesian estimates to converge on the observed win rates - the more consistently a player has demonstrated his ability to win at a particular rate, the more likely that that is his true rate. And if we recalculate the above considering a sample of 100,000 hands, we obtain:

Win Rate	$p(A\|B)$
- 5 BB/100	0.00%
- 4 BB/100	0.00%
- 3 BB/100	0.00%
- 2 BB/100	0.00%
- 1 BB/100	1.57%
0 BB/100	33.42%
- 1 BB/100	58.37%
- 2 BB/100	6.60%
- 3 BB/100	0.04%
Total	100%

By this sample size, we are much more confident that the observations match the reality, even though in the underlying distribution, only ten percent of players win at least 1 BB/100.

It is worth noting that there is somewhat of a schism between classical statisticians (sometimes called ***frequentists)*** and Bayesian statisticians. This disagreement centers (roughly) on the Bayesian idea that population parameters should be thought of as probability distributions. Bayesians estimate a prior distribution, observe a sample, and then adjust their prior distribution in light of the new evidence. Frequentists reject this approach, preferring to characterize population parameters as fixed, unchanging values, even if we cannot know their value. We are strongly oriented toward the Bayesian point of view because of its usefulness in poker analysis.

The method we have used here is not even a full-fledged Bayesian technique, but merely a simplification of techniques of Bayesian analysis intended to make the idea accessible. For more

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detail on this and on the divide between practitioners of Bayesian techniques and classical statisticians, we suggest that you consult advanced statistics texts, particularly those that have thorough treatments of point estimation and Bayesian methods.

One topic that emerges from this analysis is the idea of ***regression to the mean.*** Suppose you take an observation - perhaps of win rates over a sample that is of a small size. The idea here is that observed win rates above the mean will tend be lower if you repeat the observation in the future, and observed win rates below the mean will tend to perform better in the future. This is not because of any kind of statistical "evening out" where outlying trials from the past are offset by outlying trials (in the other direction) in the future - these events are independent. The principle that is at work here is that if you have outperformed the mean of the entire population, it is somewhat more likely that you have outperformed your expectation as well, while if you have underperformed it, you have likely underperformed your own expectation also. As a result, you will tend (sometimes very slightly) to regress a little toward the mean of all players - that is, the best prediction we can make contains a small amount of the overall mean mixed with observed results. We can see this at work in the Bayesian analysis of our hypothetical player - after 16,900 hands, his adjusted win rate prediction was still very heavily influenced by the population distribution, which dragged his win rate down toward the population mean of -1 BB/100.

Key Concepts

- When estimating population parameters from observed data, we can use one of two methods: the frequentist or classical method, or a Bayesian method.
- The classical method assumes that we have no information about the underlying distribution. This gives rise to a maximum likelihood estimate equal to the sample mean, and a confidence interval that includes all values for which the observed result is inside the significance level.
- Bayes' rule gives us a robust methodology for incorporating new information into prior probability estimates.
- Using a prior probability distribution and applying Bayes' theorem can yield better (Bayesian) estimators of population parameters when we have accurate information about the prior distribution.

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Part II: Exploitive Play

It is not enough to be a good player; you must also play well.

Siegbert Tarrasch

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